

The sharp-interface and phase-field cell problems are the same problem

1 Setup

A periodic cell Ω splits into two subdomains meeting on an interface Γ ,

$$\Omega = \Omega_1 \cup \Omega_2, \quad \Omega_1 \cap \Omega_2 = \emptyset, \quad \Gamma = \partial\Omega_1 \cap \partial\Omega_2,$$

with constant conductivities $K_1, K_2 > 0$ and $\hat{\mathbf{n}}$ the unit normal on Γ pointing out of Ω_1 . Homogenization gives the effective tensor $\mathbf{K}_{\text{eff}}$ from a *corrector* (cell) problem posed on Ω . There are two ways to write that problem, and this note shows they are the same one. Nothing below uses any property of the phases beyond K_1, K_2 being positive constants.

Notation. χ is the *indicator function* of Ω_1 : the function equal to 1 inside Ω_1 and 0 outside,

$$\chi(\mathbf{x}) = \mathbf{1}_{\Omega_1}(\mathbf{x}) = \begin{cases} 1, & \mathbf{x} \in \Omega_1, \\ 0, & \mathbf{x} \in \Omega_2. \end{cases} \quad (1)$$

φ is the phase field: a smooth function equal to 1 in Ω_1 and 0 in Ω_2 , sweeping between them across a band of decay length ε . It is a smoothed χ . $\langle \cdot \rangle = |\Omega|^{-1} \int_{\Omega}(\cdot)$, and $\llbracket \cdot \rrbracket$ is the jump across Γ .

Problem A (sharp). Find cell-periodic $\mathbf{t}$ with

$$\nabla \cdot (K_1(\nabla \mathbf{t} + \mathbf{I})) = \mathbf{0} \text{ in } \Omega_1, \quad \nabla \cdot (K_2(\nabla \mathbf{t} + \mathbf{I})) = \mathbf{0} \text{ in } \Omega_2, \quad (2)$$

$$\llbracket \mathbf{t} \rrbracket = \mathbf{0} \text{ on } \Gamma, \quad \llbracket K(\nabla \mathbf{t} + \mathbf{I}) \cdot \hat{\mathbf{n}} \rrbracket = \mathbf{0} \text{ on } \Gamma, \quad (3)$$

normalized by $\langle \mathbf{t} \rangle = \mathbf{0}$, and then

$$\mathbf{K}_{\text{eff}} = \frac{1}{|\Omega|} \left[\int_{\Omega_1} K_1(\nabla \mathbf{t} + \mathbf{I}) d\Omega + \int_{\Omega_2} K_2(\nabla \mathbf{t} + \mathbf{I}) d\Omega \right]. \quad (4)$$

The two interface conditions are the physics: the field is continuous, and the normal flux is conserved. This is Calonne et al. (2011), Eqs. (7)–(11).

Problem B (phase field). Find cell-periodic $\mathbf{t}$ with

$$\nabla \cdot (K(\varphi)(\nabla \mathbf{t} + \mathbf{I})) = \mathbf{0} \text{ in } \Omega, \quad \langle \mathbf{t} \rangle = \mathbf{0}, \quad K(\varphi) = K_2 + (K_1 - K_2)\varphi, \quad (5)$$

$$\mathbf{K}_{\text{eff}} = \frac{1}{|\Omega|} \int_{\Omega} K(\varphi)(\nabla \mathbf{t} + \mathbf{I}) d\Omega. \quad (6)$$

One domain, one equation, no interface conditions.

Claim. A and B are the same problem. Two steps: §2 collapses A's two domains into one weak equation; §3 shows A's coefficient is B's coefficient. Neither step is a limit.

2 Step 1: two domains become one weak equation

Let $v \in H_{\text{per}}^1(\Omega)$ be an arbitrary cell-periodic scalar test function. We treat each subdomain separately, then add.

(a) Test each bulk equation on its own subdomain. Multiply the first equation of (2) by v and integrate over Ω_1 ; do the same on Ω_2 :

$$\int_{\Omega_1} v \nabla \cdot (K_1(\nabla \mathbf{t} + \mathbf{I})) d\Omega = 0, \quad \int_{\Omega_2} v \nabla \cdot (K_2(\nabla \mathbf{t} + \mathbf{I})) d\Omega = 0. \quad (7)$$

Both hold because the integrands vanish pointwise. No approximation yet.

(b) Integrate by parts on each subdomain. Using $\int_D v \nabla \cdot \mathbf{A} = -\int_D \nabla v \cdot \mathbf{A} + \oint_{\partial D} v \mathbf{A} \cdot \hat{\mathbf{n}}_D$ with $\hat{\mathbf{n}}_D$ the outward normal of D . The boundary of Ω_1 consists of Γ , where its outward normal is $+\hat{\mathbf{n}}$, and whatever part of the outer cell boundary $\partial\Omega$ it touches; likewise for Ω_2 , whose outward normal on Γ is $-\hat{\mathbf{n}}$. Writing $\mathbf{A}_m = K_m(\nabla \mathbf{t} + \mathbf{I})$ for $m = 1, 2$,

$$0 = -\int_{\Omega_1} \nabla v \cdot \mathbf{A}_1 d\Omega + \int_{\Gamma} v \mathbf{A}_1 \cdot \hat{\mathbf{n}} d\Gamma + \underbrace{\int_{\partial\Omega \cap \partial\Omega_1} v \mathbf{A}_1 \cdot \hat{\mathbf{n}}_{\partial\Omega} dS}_{O_1}, \quad (8)$$

$$0 = -\int_{\Omega_2} \nabla v \cdot \mathbf{A}_2 d\Omega - \int_{\Gamma} v \mathbf{A}_2 \cdot \hat{\mathbf{n}} d\Gamma + \underbrace{\int_{\partial\Omega \cap \partial\Omega_2} v \mathbf{A}_2 \cdot \hat{\mathbf{n}}_{\partial\Omega} dS}_{O_2}. \quad (9)$$

The sign flip on the Γ term of (9) is the only subtlety: Γ is traversed with opposite orientation from the two sides.

(c) Add the two equations. This is the step that produces a single statement on Ω . Adding (8) and (9),

$$0 = -\left[\int_{\Omega_1} \nabla v \cdot \mathbf{A}_1 d\Omega + \int_{\Omega_2} \nabla v \cdot \mathbf{A}_2 d\Omega \right] + \int_{\Gamma} v (\mathbf{A}_1 - \mathbf{A}_2) \cdot \hat{\mathbf{n}} d\Gamma + O_1 + O_2. \quad (10)$$

Now dispose of the three groups in turn.

(d) The outer boundary terms cancel. $O_1 + O_2$ is an integral over all of $\partial\Omega$. Both v and the flux $K(\nabla \mathbf{t} + \mathbf{I})$ are cell-periodic, so they take equal values at corresponding points of opposite faces, where $\hat{\mathbf{n}}_{\partial\Omega}$ points in opposite directions. Every contribution is therefore matched by its negative: $O_1 + O_2 = 0$.

(e) The interface term is exactly the flux jump, which vanishes. By definition of the jump, $(\mathbf{A}_1 - \mathbf{A}_2) \cdot \hat{\mathbf{n}} = \llbracket K(\nabla \mathbf{t} + \mathbf{I}) \cdot \hat{\mathbf{n}} \rrbracket$, so the middle term of (10) is $\int_{\Gamma} v \llbracket K(\nabla \mathbf{t} + \mathbf{I}) \cdot \hat{\mathbf{n}} \rrbracket d\Gamma$, which is 0 by the second interface condition in (3) — for every v , since the integrand vanishes pointwise.

(f) The two volume integrals merge into one. This is where K_* comes from, and it is pure bookkeeping. Define the piecewise-constant coefficient

$$K_* := K_1 \chi + K_2(1 - \chi), \quad \text{so} \quad K_* = K_1 \text{ on } \Omega_1, \quad K_* = K_2 \text{ on } \Omega_2. \quad (11)$$

Because K_* equals K_1 at every point of Ω_1 , we may substitute it there without changing the integral, and likewise on Ω_2 :

$$\int_{\Omega_1} \nabla v \cdot \mathbf{A}_1 + \int_{\Omega_2} \nabla v \cdot \mathbf{A}_2 = \int_{\Omega_1} \nabla v \cdot K_*(\nabla \mathbf{t} + \mathbf{I}) + \int_{\Omega_2} \nabla v \cdot K_*(\nabla \mathbf{t} + \mathbf{I}). \quad (12)$$

The two integrands are now the *same function* of $\mathbf{x}$, integrated over two disjoint sets whose union is Ω . Additivity of the integral gives

$$\int_{\Omega_1} (\cdots) + \int_{\Omega_2} (\cdots) = \int_{\Omega_1 \cup \Omega_2} (\cdots) = \int_{\Omega} (\cdots), \quad (13)$$

the last equality because $\Omega_1 \cup \Omega_2$ differs from Ω only by Γ , a set of measure zero. So $K_\star$ is not assumed or fitted; it is the name for “ K_1 on one piece and K_2 on the other,” and (13) is the only thing done with it.

(g) Result. Substituting (d), (e), (f) into (10):

$$\boxed{\int_{\Omega} \nabla v \cdot K_\star (\nabla \mathbf{t} + \mathbf{I}) d\Omega = 0 \quad \forall v \in H_{\text{per}}^1(\Omega).} \quad (14)$$

Remark 1 (Where the two interface conditions went). *They were not dropped; each was absorbed somewhere.* Flux conservation $\llbracket K(\nabla \mathbf{t} + \mathbf{I}) \cdot \hat{\mathbf{n}} \rrbracket = 0$ was used in step (e), and it is the natural condition of (14): run (e) backwards and any weak solution satisfies it automatically. Continuity $\llbracket \mathbf{t} \rrbracket = 0$ is absorbed in the statement “ $\mathbf{t} \in H^1(\Omega)$ ”: a single H^1 field on Ω cannot jump. Had $\mathbf{t}$ been allowed to jump, (14) would not even be well posed.

Remark 2 (Why one arbitrary v on Ω is not an extra assumption). *It may look as though testing with a single v defined on all of Ω smuggles in something. It does not, and the step reverses: given (14), choose v supported strictly inside Ω_1 . Then the integral is over Ω_1 only, where $K_\star = K_1$; undoing the integration by parts (no boundary term, since v vanishes near $\partial\Omega_1$) returns the first equation of (2). Symmetrically for Ω_2 . Allowing v to straddle Γ then returns the interface term, hence (3). So (14) is equivalent to (2)–(3), not weaker.*

Remark 3 (Same algebra for the physical field). *Nothing in (a)–(g) used the specific form $\nabla \mathbf{t} + \mathbf{I}$. Replacing it by ∇T throughout collapses the physical problem $\nabla \cdot (K \nabla T) = 0$, $\llbracket T \rrbracket = \llbracket K \nabla T \cdot \hat{\mathbf{n}} \rrbracket = 0$ into $\int_{\Omega} \nabla v \cdot K_\star \nabla T = 0$ identically. See §7.*

3 Step 2: the coefficient is already the phase-field interpolation

It is tempting to finish by saying “now replace $K_\star$ by a smooth $K(\varphi)$ that converges to it.” That would be an approximation, and would need a convergence theorem. No replacement is needed.

Lemma 1. *Let $K(\cdot)$ be any function with $K(0) = K_2$ and $K(1) = K_1$. Then*

$$K(\chi) = K_\star \quad \text{pointwise on } \Omega, \quad (15)$$

exactly — no limit, no smallness of ε .

Proof. χ takes only the values 0 and 1. At $\mathbf{x} \in \Omega_1$, $\chi = 1$, so $K(\chi) = K(1) = K_1 = K_\star$. At $\mathbf{x} \in \Omega_2$, $\chi = 0$, so $K(\chi) = K(0) = K_2 = K_\star$. Those are all the points of Ω . $\square$

For the affine K of (5) it is three lines:

$$K_\star \stackrel{(11)}{=} K_1 \chi + K_2 (1 - \chi) = K_2 + (K_1 - K_2) \chi = K(\chi). \quad (16)$$

What this means. Problems A and B do not have two constitutive laws that agree asymptotically. They have *one* law $K(\cdot)$, evaluated at two different arguments:

$$\text{A: } K(\chi), \quad \text{B: } K(\varphi).$$

So Step 2 approximates the *geometry*, never the equations, and every difference between A and B traces to the single object $\varphi - \chi$. A second consequence: since χ never takes an intermediate value, the *shape* of K between 0 and 1 is invisible to Problem A. Every admissible interpolation — arithmetic, harmonic, any other — coincides exactly on χ . The choice can therefore matter only through the diffuse band (§5).

Proposition 1. *If $\varphi = \chi$, Problems A and B are the same boundary-value problem and give the same $\mathbf{K}_{\text{eff}}$, identically.*

Proof. Step 1 gives $A \Leftrightarrow (14)$. Lemma 1 gives $K_\star = K(\chi)$, so (14) is the weak form of (5). The two $\mathbf{K}_{\text{eff}}$ formulas agree by (12)–(13) applied to the integrand $K(\nabla \mathbf{t} + \mathbf{I})$. $\square$

4 Finite ε : what is left over

In practice $\varphi \neq \chi$, so Proposition 1 does not apply verbatim. By Lemma 1 the entire residual difference is carried by $\varphi - \chi$. For the affine K this is exact:

$$\delta K := K(\varphi) - K_\star = (K_1 - K_2)(\varphi - \chi), \quad \text{supp } \delta K = B_\varepsilon := \{0 < \varphi < 1\}. \quad (17)$$

Both problems are uniformly elliptic on the same bounds, $\min(K_1, K_2) \leq K \leq \max(K_1, K_2)$, so both are well posed and both admit the Dirichlet minimum principle: for a unit direction $\mathbf{e}$,

$$A(K) := \mathbf{e} \cdot \mathbf{K}_{\text{eff}}[K] \mathbf{e} = \min_{v \in H_{\text{per}}^1(\Omega)} E_K[v], \quad E_K[v] := \langle (\nabla v + \mathbf{e}) \cdot K(\nabla v + \mathbf{e}) \rangle. \quad (18)$$

Write $K_1 \rightarrow K(\varphi)$, $K_2 \rightarrow K_\star$ with minimizers $v_\varepsilon, v_\star$. For *any* v , $E_{K(\varphi)}[v] - E_{K_\star}[v] = \langle \delta K | \nabla v + \mathbf{e} |^2 \rangle$. Testing each minimizer against the other's energy gives two inequalities which combine into

$$\langle \delta K | \nabla v_\varepsilon + \mathbf{e} |^2 \rangle \leq A_\varepsilon - A_\star \leq \langle \delta K | \nabla v_\star + \mathbf{e} |^2 \rangle. \quad (19)$$

Both sides are integrals over B_ε only. The left one uses only the corrector already computed, so (19) is an *a posteriori* error bar, not an asymptotic claim. With the logistic profile $\varphi = \sigma(s/\varepsilon)$, $\sigma(u) = (1 + e^{-u})^{-1}$, in the signed distance s to Γ , and $\int_0^\infty (1 - \sigma) du = \ln 2$,

$$\|\varphi - \chi\|_{L^1(\Omega)} = 2 \ln 2 \, \varepsilon |\Gamma| + \mathcal{O}(\varepsilon^2) \implies |A_\varepsilon - A_\star| \leq 2 \ln 2 (K_1 - K_2) M^2 \frac{|\Gamma|}{|\Omega|} \varepsilon + \mathcal{O}(\varepsilon^2), \quad (20)$$

with $M^2 = \max_{B_\varepsilon} |\nabla v + \mathbf{e}|^2$. First order, computable prefactor, and no H-convergence theorem required. (That theorem is the right citation only for a non-affine K , where (17) becomes an inequality.)

5 The first-order error is a surface excess

To first order a thin graded layer is equivalent to a sharp interface carrying two *surface excesses* per unit area. Transport tangential to Γ crosses the layer's sublayers in parallel and sees an excess conductance; transport normal to it crosses them in series and sees an excess resistance:

$$\Sigma_t = \int_{-\infty}^\infty [K(\varphi) - K_\star] ds, \quad \Sigma_n = \int_{-\infty}^\infty \left[\frac{1}{K(\varphi)} - \frac{1}{K_\star} \right] ds. \quad (21)$$

These two numbers are the only $\mathcal{O}(\varepsilon)$ data the band presents. Both are elementary for the logistic profile, whose defining property here is the antisymmetry $\sigma(-u) = 1 - \sigma(u)$.

$\Sigma_t = 0$, **exactly**. K is affine, so the integrand is $(K_1 - K_2)[\sigma(u) - H(u)]$ with H the unit step, and antisymmetry makes $\sigma - H$ an odd function:

$$\Sigma_t = (K_1 - K_2) \varepsilon \int_{-\infty}^{\infty} [\sigma(u) - H(u)] du = 0. \quad (22)$$

Not first order — zero. The diffuse cell has *exactly* the sharp cell's phase fraction, so the parallel response carries no error at any ε . The leading correction is $\mathcal{O}(\varepsilon^2 \kappa)$ from interface curvature.

$\Sigma_n \neq 0$. Here $1/K$ is convex in φ , so the band short-circuits a series resistance. With $p = \sigma(u)$, $du = dp/[p(1-p)]$ and partial fractions,

$$\Sigma_n = \varepsilon \int_0^1 \left[\frac{1}{K_2 + (K_1 - K_2)p} - \frac{1}{K_*} \right] \frac{dp}{p(1-p)} = -\varepsilon \left(\frac{1}{K_2} - \frac{1}{K_1} \right) \ln \frac{K_1}{K_2} < 0. \quad (23)$$

Negative excess resistance: the diffuse cell conducts too well normal to Γ , first order in ε , with no free constants. This is the quantitative form of the familiar statement that an arithmetic mixture rule is right in parallel and wrong in series.

Corollary 1 (An interpolation with no first-order bias). *The harmonic law $K_h(\varphi) = (\varphi/K_1 + (1-\varphi)/K_2)^{-1}$ has $1/K_h$ affine in φ , so the same antisymmetry gives $\Sigma_n = 0$ — but then $\Sigma_t \neq 0$. No scalar law nulls both. The tensorial law*

$$\mathbf{K}(\varphi) = K(\varphi)(\mathbf{I} - \hat{\mathbf{n}} \otimes \hat{\mathbf{n}}) + K_h(\varphi) \hat{\mathbf{n}} \otimes \hat{\mathbf{n}}, \quad \hat{\mathbf{n}} = \nabla \varphi / |\nabla \varphi|, \quad (24)$$

uses each where its excess vanishes, giving $\Sigma_t = \Sigma_n = 0$ and $\mathcal{O}(\varepsilon^2)$. Both branches satisfy $K(0) = K_2$, $K(1) = K_1$, so Lemma 1 and Proposition 1 still apply.

This is available because the conductivity cell problem is post-processing on a frozen φ : its interpolation is decoupled from whatever the evolution model uses.

6 Worked example and test: the laminate

For layers of period L with phase fraction $\bar{\varphi}$ and n_Γ interfaces per period, $\mathbf{K}_{\text{eff}}$ is diagonal with $k_\parallel = \langle K \rangle$ and $k_\perp = \langle K^{-1} \rangle^{-1}$. Equations (22)–(23) give

$$k_\parallel(\varepsilon) = K_2 + (K_1 - K_2)\bar{\varphi} \quad \text{exact for every } \varepsilon, \quad (25)$$

$$\langle K^{-1} \rangle_\varepsilon = \frac{\bar{\varphi}}{K_1} + \frac{1 - \bar{\varphi}}{K_2} - \frac{n_\Gamma \varepsilon}{L} \left(\frac{1}{K_2} - \frac{1}{K_1} \right) \ln \frac{K_1}{K_2}, \quad k_\perp = \langle K^{-1} \rangle_\varepsilon^{-1}. \quad (26)$$

Equation (26) is not a truncation: its only error is tail overlap between neighbouring interfaces, $\mathcal{O}(e^{-L/4\varepsilon})$, which against direct quadrature is 5×10^{-5} relative at $\varepsilon = L/50$ and machine precision by $\varepsilon = L/128$.

In our application $K_1/K_2 = 2.29/0.02 = 114.5$, giving $(K_2^{-1} - K_1^{-1}) \ln(K_1/K_2) = 234.959$ and, at $\bar{\varphi} = \frac{1}{2}$, $n_\Gamma = 2$:

ε	$\langle K^{-1} \rangle$	$k_{\perp}$	bias
$L/50$	15.8200	0.063211	+59.4%
$L/128$	21.5471	0.046410	+17.0%
$L/512$	24.3005	0.041151	+3.8%
sharp	25.2183	0.039654	—

How to test it. Equation (26) is *linear in ε in resistivity* and a hyperbola in conductivity. Fit and plot $1/k_{\perp}$ against ε ; then the test has two predicted numbers and no fitted ones,

$$\text{intercept} = \frac{\bar{\varphi}}{K_1} + \frac{1 - \bar{\varphi}}{K_2}, \quad \text{slope} = \frac{n_{\Gamma}}{L} \left(\frac{1}{K_2} - \frac{1}{K_1} \right) \ln \frac{K_1}{K_2}. \quad (27)$$

Two ways to get a meaningless answer: evaluating (27) at the nominal rather than the *measured* $\bar{\varphi}$ (the discrete field's mean is not exactly $\bar{\varphi}$), and refining ε at fixed mesh, which confounds this bias with mesh convergence. The committed driver is `studies/keff_sharp_limit/verification/`.

7 Carrying this to the transient equation with a phase-change source

The evolution model solves, instead of (5), an energy balance of the form

$$\underbrace{\rho(\varphi)c(\varphi)}_{\text{storage}} \partial_t T = \underbrace{\nabla \cdot (K(\varphi) \nabla T)}_{\text{conduction}} + \underbrace{\rho_1 L \partial_t \varphi}_{\text{latent heat}}. \quad (28)$$

Three things change, and they are not equally benign.

1. Conduction: nothing new. The operator is the same $K(\varphi)$ appearing in (5). Step 1 collapses the sharp two-domain heat equation in the same way, Lemma 1 applies unchanged, and (22)–(23) hold verbatim. So the evolution model carries the *same* first-order conduction bias — too conductive normal to Γ — and Corollary 1 is the same fix.

2. Storage: a nonzero excess, from a quadratic in φ . Repeating (21) for the volumetric heat capacity $C = \rho c$, the excess stored enthalpy per unit interface area is $\Sigma_C = \int [C(\varphi) - C_{\star}] ds$. If C is interpolated *affinely* in φ this vanishes exactly, by the same antisymmetry as (22). But interpolating ρ and c *separately* and multiplying, as (28) is written, makes C *quadratic*:

$$\rho(\varphi)c(\varphi) = \rho_2 c_2 + (\rho_2 \Delta c + c_2 \Delta \rho) \varphi + \Delta \rho \Delta c \varphi^2, \quad \Delta \rho = \rho_1 - \rho_2, \quad \Delta c = c_1 - c_2, \quad (29)$$

and φ^2 is *not* antisymmetric about the interface. Its excess is exactly computable by the substitution used for (23):

$$\int_{-\infty}^{\infty} [\sigma(u)^2 - H(u)] du = -1 \quad \implies \quad \Sigma_C = -\varepsilon \Delta \rho \Delta c \neq 0. \quad (30)$$

The diffuse cell stores *less* enthalpy than the sharp one, at first order in ε and proportional to the interfacial area. With $\rho_1 c_1 = 1.80 \times 10^6$ and $\rho_2 c_2 = 1.40 \times 10^3 \text{ J m}^{-3} \text{ K}^{-1}$, $\Delta \rho \Delta c = 8.41 \times 10^5$, so each interface loses the heat capacity of a 0.47ε -thick slab of phase 1. On a cell with specific surface area $|\Gamma|/|\Omega| = 2 \times 10^4 \text{ m}^{-1}$ that is a 1.9% deficit at $\varepsilon = 1 \mu\text{m}$ and 9.3% at $5 \mu\text{m}$. *The fix is free*: interpolate the product, $C(\varphi) = \rho_2 c_2 + (\rho_1 c_1 - \rho_2 c_2) \varphi$, which is affine and has $\Sigma_C = 0$ exactly. The two agree at $\varphi \in \{0, 1\}$, so Lemma 1 is indifferent between them — this is precisely a case where the interpolation is invisible to the sharp problem and matters only through the band.

3. Latent heat: the integral is exact, the placement is not. In the sharp model latent heat is a *surface* source on Γ , entering as a Stefan jump condition $[[K\nabla T \cdot \hat{\mathbf{n}}]] = \rho_1 L v_n$. The phase-field model spreads it over the band as the volumetric $\rho_1 L \partial_t \varphi$. The total released is right:

$$\int_{\Omega} \rho_1 L \partial_t \varphi d\Omega = \rho_1 L \frac{d}{dt} \int_{\Omega} \varphi d\Omega = \rho_1 L \frac{dV_1}{dt}, \quad (31)$$

exactly, because $\int_{\Omega} \varphi d\Omega$ is the exact phase-1 volume — the same $\Sigma_t = 0$ antisymmetry as (22). What is *not* exact is where within the band it is deposited. Heat is injected across a layer of width $\mathcal{O}(\varepsilon)$ over which T and K both vary strongly (here by a factor 114), so it enters at the wrong local temperature.

This third item is the one that needs care, and it is outside the scope of the analysis above: (19) and (21) assume a *frozen* φ and *no* source, which is exactly the situation of the $\mathbf{K}_{\text{eff}}$ cell problem and exactly not the situation of (28). A moving diffuse interface carrying a source is the regime of the thin-interface analyses of Karma & Rappel and Almgren, whose known $\mathcal{O}(\varepsilon)$ artifacts — modified interface kinetics, solute/thermal trapping, and a spurious surface diffusion along Γ — are generated by the source term and are most pronounced when the two phases' transport coefficients are very unequal, as here. Quantifying them requires a matched asymptotic expansion of (28), not the equilibrium excesses of §5.

Remark 4 (The practical consequence). *A clean ε ladder for $\mathbf{K}_{\text{eff}}$ certifies the conduction operator and the geometry representation. It does not certify the coupled model, which has two further ε -sensitivities: one cheap to remove ((30), by interpolating pc instead of ρ and c), and one requiring its own analysis (the latent-heat source). Conversely, item 2 suggests its own test: the total enthalpy of a static two-phase cell should be ε -independent, which is the storage analogue of the $k_{\parallel}$ gate and is equally cheap to run.*